

Dooyoung Kim

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CURRENT APPOINTMENT

From Jun 2026 **Lecturer in Computer Science and Information Technology**,
Continuing academic appointment in research and teaching.
School of Computing, Engineering and Mathematical Sciences,
La Trobe University, Melbourne, Australia.

RESEARCH INTERESTS

Areas Human–computer interaction; spatial computing and extended reality; human–AI interaction; remote collaboration and telepresence.
Research focus Spatial interfaces and representations for sharing human experiences and expertise across space and time.

EDUCATION

Mar 2021 – Feb 2024 **Ph.D. in Culture Technology**, KAIST, Daejeon, Korea.
Thesis: *Space-Adaptive Mutual Space Generation for Mixed Reality Remote Collaboration*.
UVR Lab. · Advisor: Prof. Woontack Woo.
Mar 2019 – Feb 2021 **M.S. in Culture Technology**, KAIST, Daejeon, Korea.
Thesis: *Adjusting Relative Translation Gains According to Space Size in Redirected Walking for Mixed Reality Mutual Space Generation*.
UVR Lab. · Advisor: Prof. Woontack Woo.
Visiting Researcher (Dec 2019 – Feb 2020): Imagine Lab, École des Ponts ParisTech, France (Prof. Vincent Lepetit).
Mar 2014 – Feb 2019 **B.S. in Mechanical Engineering**, KAIST, Daejeon, Korea.
Exchange Student (2016): Tsinghua University, Beijing, China.
Lab Interns: Hubo Lab — Humanoid Robotics (2017); UVR Lab — AR/VR (2018).

PREVIOUS APPOINTMENTS

Mar 2024 – May 2026 **Senior Researcher**, KAIST, Daejeon, Korea.
Co-PI on Meta-Object (US\$3M total project funding); contributed to proposal preparation and grant acquisition, with decision authority over research-equipment purchases. System Integration Manager on OpenXR (US\$6M total project funding); integrated research outputs across multiple KAIST teams. Co-advised and mentored 7 graduate students (PhD/MS), with publications at IEEE TVCG, ISMAR, VR.
Jan 2025 – Dec 2025 **NYU–KAIST Postdoc Collaboration**, New York University, NY, USA.
NYU Future Reality Lab (Prof. Ken Perlin); on-site and remote visits throughout 2025 on the KAIST–NYU Meta-Museum initiative.
May 2022 – Mar 2023 **AR/VR Researcher**, Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea.
Joint research with KAIST on the OpenXR: Untact Realistic Platform Technology government project.
Apr 2021 – Nov 2022 **Creator**, ZER01NE program, Hyundai Motors Group (HMG).
Selected as AR/VR researcher. Projects: *Automatic SONATA* — scenery style-transfer system for autonomous vehicles (2021); *Holo-bot* — XR social interface robot with holographic avatar (2022).

TECHNICAL EXPERTISE

Platforms Meta Quest 3/Pro, HoloLens 2, Apple Vision Pro, custom CAVE/wall-display setups, OpenXR.

Software	Unity (C#), Unreal, ARKit/ARCore, MRTK; Python (PyTorch, NumPy); OpenCV, COLMAP/SLAM; human-subject statistics.
Methods	XR prototyping, spatial AI / scene understanding, semantic scene graphs, 3D reconstruction, world-model priors, redirected walking, perceptual experiments, IRB-approved human-subject studies, mixed-methods evaluation.

AWARDS AND HONOURS

Oct 2025	Best Paper Award (Top 1%), IEEE ISMAR 2025 — <i>Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display</i> .
Oct 2025	Best Paper Award (Top 1%), IEEE ISMAR 2025 — <i>Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception</i> .
Jan 2025	NYU–KAIST Postdoc Collaboration Grant, New York University.
Oct 2024	Best Paper Award (Top 1%), IEEE ISMAR 2024 — <i>Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence</i> .
Mar 2023	KAIST Start-up Scholarship, KAIST Alumni Academic Scholarship Foundation.
Dec 2022	Best Presentation Award (1st Prize), The Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR) 2022 — Springer <i>Virtual Reality</i> paper invited and presented.
Apr 2022	NCSOFT Ph.D. Scholarship.
May 2021	Honorable Mention Award (Top 5%), ACM CHI 2021.

GRANTS AND RESEARCH PROJECTS

Jan 2025 – Dec 2025	Meta-Museum — NYU–KAIST Collaborative Research Grant. <i>Tele-experience system for a meta-museum between Daejeon (Korea) and New York (USA)</i> . Role: Project Leader. Collaboration with NYU Future Reality Lab (Prof. Ken Perlin).
Apr 2024 – May 2026	Meta-Object — Korean Government (IITP) national research project. <i>Real-time XR interface technology for environmental adaptation; object modelling, multimodal feedback, intelligent platform</i> . Project period: Mar 2024 – Dec 2026. Role: Co-Principal Investigator (Co-PI; PI: Prof. Woontack Woo). Contributed to proposal preparation and grant acquisition; exercised decision authority over research-equipment purchases. Directed 40 researchers across 8 multidisciplinary teams. Total project funding: US\$3M.
Dec 2021 – May 2026	OpenXR: TranSpace 3.0 — Korean Government (NST) national research project. <i>Development of untact realistic OpenXR remote collaboration platform technology</i> . Role: System Integration Manager; integration lead for research outputs across multiple KAIST teams. Total project funding: US\$6M over 6 years.
Sep 2019 – Dec 2020	TranSpace 2.0 — Korean Government (NRF) national research project. <i>User restoration technology for ultra-realistic remote virtual interaction</i> . Role: Research Assistant. AR/VR remote-collaboration system and VR locomotion research.
Apr 2020 – Dec 2020	National R&D Real Challenge — Korean Government (KIRD). <i>Reinforcement of visual cognitive abilities for AR/VR/MR remote collaboration</i> . Role: Research Assistant. Visual-perception research for VR remote-collaboration scenarios.

PUBLICATIONS

Summary	31 refereed publications in total — 12 journal ([J.1–J.12]), 9 conference ([C.1–C.9]), and 10 workshop / adjunct / demo ([W.1–W.10]). All listed below are peer-reviewed.
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Bold name indicates author; ★ marks award-winning papers. Co-advised students' first-author papers are labelled (co-advised, student first author).

Refereed journal papers

- [J.12] S.-Y. Kim, **Dooyoung Kim**, T. Son, Y. B. Kim, W. Cho, W. Woo. “Mix3R: Zero-shot Interactive 3D Scene Reconstruction from a Single Image in Mixed Reality.” *IEEE TVCG*, 2026 (accepted). (co-advised, student first author)
- [J.11] S. Kim, **Dooyoung Kim**, S. Choi, W. Woo. “Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.10] S.-Y. Kim, **Dooyoung Kim**, W. Cho, H. Song, S. Kang, W. Woo. “SceneLinker: Compositional 3D Scene Generation via Semantic Scene Graph from RGB Sequences.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.9] S. Choi, **Dooyoung Kim**, T. Ha, S. Kim, W. Woo. “Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.8] T. Ha, W. Cho, **Dooyoung Kim**, W. Woo. “Int3DNet: Scene-Motion Cross Attention Network for 3D Intention Prediction in Mixed Reality.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.7] **Dooyoung Kim**, J. Hong, H. Ko, W. Woo. “Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display.” *IEEE TVCG*, 2025. ★ Best Paper Award (Top 1%), IEEE ISMAR 2025
- [J.6] M. Baeck, Y. Shin, **Dooyoung Kim**, H. Lee, S. H. Yoon, W. Woo. “Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception.” *IEEE TVCG*, 2025. (co-advised, student first author) ★ Best Paper Award (Top 1%), IEEE ISMAR 2025
- [J.5] **Dooyoung Kim**, T. Ha, J. Hong, S. Kim, S. Choi, H. Ko, W. Woo. “Meta-Objects: Interactive and Multisensory Virtual Objects Learned from the Real World for Use in Augmented Reality.” *IEEE Computer Graphics and Applications*, 2025.
- [J.4] **Dooyoung Kim**, S. Kim, J.-e. Shin, B. Yoon, J. Kim, J. Lee, W. Woo. “The Effects of Spatial Configuration on Relative Translation Gain Thresholds in Redirected Walking.” *Springer Virtual Reality*, 2023. ★ Best Presentation Award (1st Prize), APMAR 2022
- [J.3] J. Kim, H. Jang, **Dooyoung Kim**, J. Lee. “Exploration of the VR Teleportation Methods Using Hand Tracking, Eye Tracking, and EEG.” *IJHCI*, 2023.
- [J.2] B. Yoon, J.-e. Shin, H.-i. Kim, S. Oh, **Dooyoung Kim**, W. Woo. “Effects of Avatar Transparency on Social Presence in Task-Centric Mixed Reality Remote Collaboration.” *IEEE TVCG*, 2023.
- [J.1] J.-e. Shin, B. Yoon, **Dooyoung Kim**, W. Woo. “The Effects of Spatial Complexity on Narrative Experience in Space-Adaptive AR Storytelling.” *IEEE TVCG*, 2022.

Refereed conference papers

- [C.9] S.-Y. Kim, A. Elskhawy, T. Ha, **Dooyoung Kim**, E. Shin, B. Busam, W. Woo. “DeWorldSG: Depth-Aware 3D Semantic Scene Graph Generation via World-Model Priors.” *ECCV*, 2026 (accepted). (co-advised, student first author)
- [C.8] S. Choi, S. A. Bae, **Dooyoung Kim**, W. Woo. “Object-Driven Event Retrieval with Assembled Object Interface for Mixed-Reality Assembly Playback.” *IEEE ISMAR*, 2026 (accepted). (co-advised, student first author)
- [C.7] **Dooyoung Kim**, S. Kim, S. Choi, W. Woo. “Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence.” *IEEE ISMAR*, 2024. ★ Best Paper Award (Top 1%)
- [C.6] S. Kim, **Dooyoung Kim**, J.-e. Shin, W. Woo. “Object Cluster Registration of Dissimilar Rooms using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces.” *IEEE VR*, 2024. (co-advised, student first author)
- [C.5] **Dooyoung Kim**, W. Woo. “Edge-Centric Space Rescaling with Redirected Walking for Dissimilar Physical–Virtual Space Registration.” *IEEE ISMAR*, 2023.

- [C.4] **Dooyoung Kim**, J. Kim, J.-e. Shin, B. Yoon, J. Lee, W. Woo. “Effects of Virtual Room Size and Objects on Relative Translation Gain Thresholds in Redirected Walking.” *IEEE VR*, 2022.
- [C.3] J.-e. Shin, B. Yoon, **Dooyoung Kim**, H.-i. Kim, W. Woo. “The Effects of Device and Spatial Layout on Social Presence during a Dynamic Remote Collaboration Task in Mixed Reality.” *IEEE ISMAR*, 2022.
- [C.2] **Dooyoung Kim**, J.-e. Shin, J. Lee, W. Woo. “Adjusting Relative Translation Gains According to Space Size in Redirected Walking for Mixed Reality Mutual Space Generation.” *IEEE VR*, 2021.
- [C.1] J.-e. Shin, B. Yoon, **Dooyoung Kim**, H.-i. Kim, W. Woo. “A User-Oriented Approach to Space-Adaptive Augmentation: The Effects of Spatial Affordance on Narrative Experience in an Augmented Reality Detective Game.” *ACM CHI*, 2021. ★ Honorable Mention Award (Top 5%)

Refereed workshop, adjunct, and demo papers

- [W.10] **Dooyoung Kim**, W. Woo. “Spatio-Temporal Mixed and Augmented Reality Experience Description for Interactive Playback.” *IEEE ISMAR-Adjunct*, 2025.
- [W.9] **Dooyoung Kim**, K. Kim, C. T. Silva, Q. Sun, D. Turakhia, Z. Wang, K. Wang, K. Perlin, S. Feiner, W. Woo. “An Overview of the 1st International Workshop on Spatial Memory in XR: The Future of Memory Capture and Replay Through XR and AI (XRMemory).” *IEEE VRW*, 2025.
- [W.8] S. Kim, **Dooyoung Kim**, T. Son, W. Woo. “RealityCrafter: User-Guided Editable 3D Scene Generation from a Single Image in Mixed Reality.” *ACM UIST-Adjunct*, 2025. (*co-advised, student first author*)
- [W.7] T. Ha, S. Choi, S. Kim, **Dooyoung Kim**, W. Woo. “Human–Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph.” *IEEE VRW*, 2025. (*co-advised, student first author*)
- [W.6] J. Kim, **Dooyoung Kim**, B. Kim, H. Kim, J. Lee. “Holobot: Hologram-Based Extended Reality Telepresence Robot.” *ACM/IEEE HRI Companion*, 2023.
- [W.5] **Dooyoung Kim**, H.-i. Kim, W. Woo. “Mutual Space Generation with Relative Translation Gains in Redirected Walking for Asymmetric Remote Collaboration.” *IEEE ISMAR-Adjunct*, 2022.
- [W.4] J. Du, S. Park, J. Yum, Z. Ö. Özdemir, **Dooyoung Kim**, S. Y. Oh, S. H. Yoon. “Art Rich: Place Your AR Artwork.” *IEEE ISMAR-Adjunct*, 2022.
- [W.3] S. Kim, H. Jang, K. T. Oh, S. Y. Oh, **Dooyoung Kim**, W. Woo, J. Lee, J. Ahn, S. H. Yoon. “Bring Store in My Room: AR Store Authoring System for Spatial Experience in Mobile Shopping.” *IEEE ISMAR-Adjunct*, 2022.
- [W.2] S. U. Lee, J. Yun, D. Kim, **Dooyoung Kim**, S. Y. Oh, S. H. Yoon. “CARDS: Comprehensive AR Docent System.” *IEEE ISMAR-Adjunct*, 2022.
- [W.1] H.-I. Kim, T. Kim, E. Song, S. Y. Oh, **Dooyoung Kim**, W. Woo. “Multi-Scale Mixed Reality Collaboration for Digital Twin.” *IEEE ISMAR-Adjunct*, 2021.

PATENTS AND STANDARDISATION

Granted patents

- 2026 [US] Edge-centric space rescaling method for dissimilar space registration and system therefor (US12555342B2).
- 2025 [US/KR] Virtual reality space adjusting method with relative translation gain in redirected walking and the system thereof (US12333628B2; KR102829349B1).
- 2024 [KR] Method and system for generating remote collaboration mutual space (KR102728698B1).

Filed patent applications

Years indicate filing dates.

- 2026 [KR] Method and system for processing mixed and augmented reality experiences. *Filed; not yet published.*

- 2026 [KR] Image processing device and recording medium for storing video processed therefrom. *Filed; not yet published.*
- 2026 [KR] A system for real-time affordance visualization and interaction with meta-objects based on dynamic user context. *Filed; not yet published.*
- 2024 [KR] Method of operating mixed reality telepresence system and mixed reality telepresence system performing the same (app. 10-2024-0005235; pub. KR20250110967A).
- 2024 [KR] Spatial extension using human stereoscopic perception with a large wall display (app. 10-2024-0184973).
- 2024 [KR] Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces (app. 10-2024-0165710).
- 2023 [KR] Edge-centric space rescaling method for dissimilar space registration and the system thereof (app. 10-2023-0100226; pub. KR20250019288A).

Published PCT applications

Years indicate filing dates.

- 2025 [PCT] Method for operating mixed reality telepresence system and mixed reality telepresence system performing same (PCT/KR2025/000529; pub. WO2025150937A1).
- 2023 [PCT] Method and system for generating shared space for remote collaboration (PCT/KR2023/016052; pub. WO2024219569A1).

International standardisation

- 2025 ISO/IEC JTC 1/SC 24 — *Spatio-Temporal Mixed and Augmented Reality Experience Description (MAR-ED) for Adaptive Playback.*

INVITED TALKS

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| July 2026 | Monash University (host: Prof. Tim Dwyer; Department of Human Centred Computing) — Connecting Intelligences beyond Boundaries. |
| May 2026 | University of Adelaide (host: APrf. Gun Lee; Australian Research Centre for Interactive and Virtual Environments) — Interactive XR Intelligence. |
| May 2026 | Handong Global University (host: Prof. Minkyu Ahn; School of Artificial Intelligence, Computer and Electrical Engineering) — XRMemory and Physical AI. |
| 2024 | Korea Computer Congress (KCC), Top-Conference Session. |
| 2024 | University of Minnesota (host: Prof. Victoria Interrante) — Redirected Walking & Spatial AI. |
| 2023 | Korea Computer Congress (KCC), Top-Conference Session. |
| 2023 | Korea Software Congress (KSC), Top-Conference Session. |
| 2023 | HCI Korea Conference — Metaverse Research and Development Session. |
| 2022 | ACM SIGGRAPH Asia — OpenXR BoF Session. |
| 2022 | Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR), Pitch-Your-Work Session. |

CONFERENCE AND EDITORIAL SERVICE

Editorial

- 2025– Lead Guest Editor, Special Issue on “XRMemory: The Future of Memory Capture, Reconstruction, and Replay in XR,” Springer *Virtual Reality Journal* (2025 JIF: 5.4).
- 2025– Guest Editor, Special Issue on “XR for Knowledge Work and Workplace Enhancement: Human-Centered, Productive, and Ethical XR Workspaces,” Springer *Journal of Ambient Intelligence and Humanized Computing*.

Organising committee — IEEE ISMAR / VR (flagship XR venues)

- 2027 Doctoral Consortium Chair, IEEE VR 2027.
- 2026 Industrial Exhibition & Sponsorship Chair, IEEE ISMAR 2026.
- 2026 Committee Member, XR/VR Standardization Committee, IEEE VR 2026.
- 2026 Principal Organiser, 4th International XRMemory Workshop @ IEEE ISMAR 2026.
- 2026 Principal Organiser, 3rd International XRMemory Workshop @ IEEE VR 2026.
- 2025 Assistant to General Chair & Social Event Chair, IEEE ISMAR 2025 Organising Committee.
- 2025 Principal Organiser, 2nd International XRMemory Workshop @ IEEE ISMAR 2025.
- 2025 Organiser, XRWorks'25 Workshop @ IEEE ISMAR 2025; IPC, XRStand Workshop.
- 2025 Principal Organiser, 1st International XRMemory Workshop @ IEEE VR 2025.

Organising committee — other

- 2025 Organiser & Program Chair, 3rd International Symposium on Meta-AI for X (IS-MAX).
- 2025 Organiser, MetaMuseum Workshop @ HCI Korea Conference.
- 2024 Poster Chair (Computer Graphics & Interaction); Metaverse+AI+XR Workshop Organiser — Korea Computer Congress (KCC).
- 2024 Poster Chair (CG/HCI/AI/Metaverse/AR/VR) — Korea Software Congress (KSC).
- 2024 Organiser & Web Chair, 2nd International Symposium on Meta-AI for X (ISMAX).
- 2023 Organiser & Student Chair, 1st International Symposium on Meta-AI for X (IS-MAX).

International Program Committee (IPC) & Reviewer

IEEE VR	2027 (IPC), 2026 (IPC), 2025 (IPC), 2024, 2023, 2022.
IEEE ISMAR	2026 (IPC), 2025 (IPC), 2024, 2023, 2022, 2021.
ACM VRST	2026 (IPC), 2025 (IPC), 2024 (IPC).
ACM CHI	2026, 2025, 2024, 2023.
IEEE/ACM HRI	2023.

TEACHING

Lecturer — La Trobe

- 2026 CSE2002 — Agentic AI Software Engineering; developing a new course.

Teaching assistant — KAIST

- 2025 ARP700 — AR project mentoring; teaching assistant on Spatial AI & XRMemory.
- 2024 ARP700 — AR project mentoring; teaching assistant on XR & Spatial Computing.
- 2024 MV600 — AR/Metaverse project mentoring; teaching assistant on MR telepresence.
- 2023 CTP445 — AR fundamentals; discussion sessions; teaching assistant on SLAM with OpenCV.
- 2022 GCT555 — Unity & AR/VR for graduate students; teaching assistant on HoloLens 2, MRTK with Unity.
- 2022 CTP445 — Unity & AR/VR for graduate students; teaching assistant on smart-phone AR with ARCore and Unity.

RESEARCH SUPERVISION

- 2026 La Trobe University: primary supervisor for 2 master's students; co-supervisor for 1 PhD student and 1 master's student.
- Mar 2024 – May 2026 KAIST: Co-advised and mentored 7 graduate students (PhD/MS) during postdoctoral research. Related student-first-author papers are marked in Publications.

COMMUNITY AND OUTREACH

- Jan 2023 – May 2026 Global Shaper, World Economic Forum. Founding Curator of Daejeon Hub; Vice Curator (2023–2024).
- Mar 2022 – Dec 2022 Instructor, *Multi-User Game Development with Unity*, KAIST Center for Gifted Education.
- Jul 2021 – Aug 2021 Python Coach, NAVER.